

A Deep Learning Model Integrating Transformer and MambaBlock Modules with Bi-LSTM for Power Load Forecasting Considering Holiday Effects

Siyang Ma

Sino-European School of Technology of Shanghai
University (UTSEUS),
Shanghai University,
Shanghai, China
masiyang66666@163.com

Zhaohui Ning*

School of Mechatronic and Electrical Engineering and
Automation,
Shanghai University,
Shanghai, China
*ningzhaohui666@163.com

Abstract—Accurate power load forecasting plays a pivotal role in the management and operation of power systems, directly contributing to grid stability and optimal resource utilization. This study introduces an advanced deep learning model that combines Transformer encoders, MambaBlock modules, Bidirectional Long Short-Term Memory (Bi-LSTM) networks, and multi-head attention mechanisms. The model is specifically designed to incorporate the effects of Chinese holidays and compensatory workdays on power demand. By employing extensive feature engineering and a customized Weighted Mean Squared Error (Weighted MSE) loss function, the proposed model achieves outstanding forecasting accuracy. On Region A's 2014 dataset, it delivers a mean absolute error (MAE) of 160.20 MW, a root mean squared error (RMSE) of 221.75 MW, and a mean absolute percentage error (MAPE) of 2.10%. When applied to a separate dataset from Region B, the model maintained strong performance, with a MAE of 190.39 MW, RMSE of 318.44 MW, and MAPE of 3.45%. These results highlight the model's effectiveness in addressing complex time series patterns and region-specific factors, providing both practical significance and a valuable theoretical contribution to the field of power load forecasting.

Keywords—Power Load Forecasting, Deep Learning, Transformer, MambaBlock, Bi-LSTM, Holiday Effects

I. INTRODUCTION

With the continuous growth of global energy demand and the increasing complexity of power systems, power load prediction plays a key role in the planning, dispatch and management of power systems [1], [2]. Accurate load forecasting not only optimizes electrical resource allocation and reduces operational costs, but also improves the stability and reliability of the power grid. However, power loads exhibit pronounced time series characteristics and are influenced by weather conditions, economic activities, and holidays [3], [4]. Among these, the effects of holidays and compensatory working days on power loads are particularly significant, as electricity consumption patterns during these special periods often differ markedly from those on regular days [5].

Existing power load forecasting methods mainly include traditional statistical techniques (e.g., Autoregressive Integrated Moving Average models, ARIMA), machine learning approaches (e.g., Support Vector Machines,

Random Forests), and recently emerging deep learning techniques [6]. Although deep learning methods excel at capturing complex non-linear relationships and handling large-scale data, they still face challenges when accounting for special factors such as holidays [7]. This paper aims to introduce an advanced deep learning architecture combined with comprehensive feature engineering to develop a power load forecasting model that effectively incorporates holiday effects.

The main contributions of this paper are:

- 1) Feature Engineering: Various temporal and holiday features designed for Chinese holidays and compensatory working days, including load patterns before and after holidays.
- 2) Model Architecture: Proposed a deep learning model that integrates Transformer encoders with MambaBlock modules, combined with Bidirectional Long Short-Term Memory (Bi-LSTM) networks and Multi-head Attention Mechanisms, to fully capture both global and local features of time series data.
- 3) Loss Function: Customized a Weighted Mean Squared Error (Weighted MSE) loss function to focus more on the accuracy of predictions during critical dates such as holidays.
- 4) Experimental Validation: Extensive experiments on real-world datasets were conducted to validate the effectiveness and demonstrate the superior performance of the proposed model in power load forecasting.

II. DATA ANALYSIS

A. Power Load Analysis

TABLE I DATASET VARIABLES

Variable Name	Description
Maximum Temperature (°C)	Daily maximum temperature
Minimum Temperature (°C)	Daily minimum temperature
Average Temperature (°C)	Daily average temperature
Relative Humidity (%)	Daily average relative humidity
Rainfall (mm)	Daily rainfall
Daily Average Load (MW)	Daily average power load

This study employs daily power load data from a

specific Region A spanning from January 1, 2012 to December 31, 2014. As summarized in Table I, the dataset encompasses the following variables.

B. Factor Influence Analysis

Power loads are influenced by a variety of factors, primarily including:

1) Weather Conditions: Temperature and humidity are crucial weather factors affecting power loads[8]. High or low temperatures increase the use of air conditioning or heating devices, thereby elevating power loads [9].

2) Temporal Features: These include season, month, day of the week, etc. Electricity demand varies significantly across seasons and months, and consumption patterns differ between weekdays and weekends.

3) Holiday Effects: Holidays and compensatory working days have a pronounced impact on power loads. Chinese holidays encompass traditional festivals (for example, the Spring Festival, the Mid-Autumn Festival) and working weekends formed through compensatory arrangements, which have substantial effects on power loads [10].

To effectively capture these factors, this study incorporates comprehensive temporal and holiday features during feature engineering. Additionally, it constructs load pattern features for pre- and post-holidays to reflect both long-term and short-term impacts on power loads.

III. PREDICTION ALGORITHMS RESEARCH

A. Data Preprocessing

Before model training, comprehensive data preprocessing and feature engineering were performed, including the following steps:

1) Date Processing: Converted the date column in the raw data to datetime format and generated multiple time-related features such as year, month, day, day of the week, and season.

2) Holiday Features: Utilized predefined holiday and compensatory working day dictionaries to construct features indicating whether a day is a holiday or compensatory working day, the duration of holidays, and the type of holiday.

3) Load Pattern Features Before and After Holidays: Calculated the number of days to the Spring Festival and holiday cycles to capture load variation trends before and after holidays.

4) Lag Features and Rolling Statistics: Generated lag features from 1 to 21 days and rolling mean and standard deviation features for 7, 14, and 30 days to enhance the model's memory capabilities for time series data.

5) Interaction Features: Constructed interaction features between average temperature and holidays to reflect the combined effect of temperature on holiday loads.

6) Data Cleaning: Removed rows containing missing values to ensure data integrity.

7) Feature Scaling: Applied StandardScaler for feature standardization to improve model training efficiency and stability.

B. Mathematical Modeling Algorithms

(1) Model Architecture

The proposed forecasting model integrates Transformer encoders, MambaBlock modules, Bidirectional Long Short-Term Memory (Bi-LSTM) networks, and multi-head attention mechanisms[11]. The specific architecture is illustrated in Figure 1.

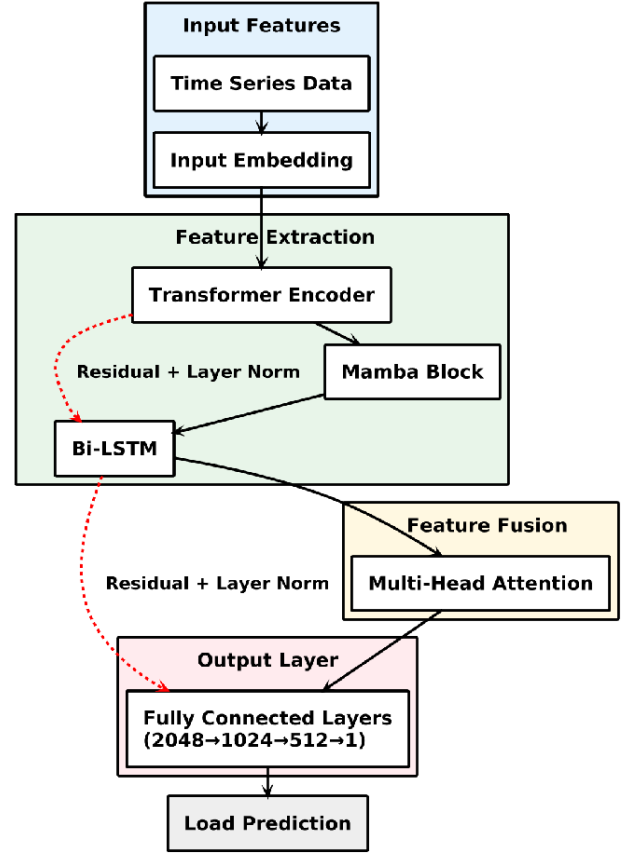


Figure 1: Model Architecture Diagram

(2) Transformer Encoder

The Transformer model, initially proposed by Vaswani et al.[12], has achieved significant success in the field of natural language processing due to its parallel computation capabilities and ability to capture long-range dependencies. The core component of the Transformer is the multi-head self-attention mechanism, defined by the following equation:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Where Q , K , and V represent Query, Key, and Value matrices, respectively, and d_k is the dimension of the keys. The multi-head attention mechanism captures information from different subspaces through multiple parallel attention heads, thereby enhancing the model's representational capacity.

In this model, the Transformer encoder is employed to extract global features from the input time series, aiding the model in understanding long-term dependencies.

(3) MambaBlock Module

MambaBlock is a custom neural network module

developed in this study, integrating convolutional layers with State Space Models (SSM)[13]. Convolutional layers are utilized to extract local features, while the state space model captures temporal dependencies through parameterized dynamic systems. The primary computational steps of the MambaBlock include input expansion, convolution operations, state updates, and output projection, as detailed below:

1) Input Expansion :

Linearly expands the input features to increase the model's representational capacity.

$$x_{\text{expanded}} = W_{\text{expand}}x + b_{\text{expand}} \quad (2)$$

2) Convolution Operation :

Applies one-dimensional convolution to extract local features.

$$x_{\text{conv}} = \text{Conv1D}(x_{\text{expanded}}) \quad (3)$$

3) State Update :

Updates the hidden state using the state space model[13].

$$h_t = h_{t-1} + u_t B \quad (4)$$

$$y_t = h_t C^T + u_t D \quad (5)$$

Where h_t is the hidden state at time step t , u_t is the input, and B, C, D are trainable parameters.

4) Output Projection:

Projects the output of the state space model back to the original dimension.

$$y_{\text{projected}} = W_{\text{project}}y + b_{\text{project}} \quad (6)$$

Through these steps, the MambaBlock effectively captures both local and global features within the time series data[13].

(4) Bi-LSTM and Attention Mechanism

Bidirectional Long Short-Term Memory (Bi-LSTM) networks enhance the model's ability to capture sequential information by processing the data in both forward and backward directions. The fundamental computational units of Bi-LSTM include the input gate, forget gate, and output gate, with the update equations as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (7)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (8)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (9)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (10)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (11)$$

$$h_t = o_t * \tanh(C_t) \quad (12)$$

Where σ denotes the sigmoid activation function, and $*$ represents element-wise multiplication [14].

The Bi-LSTM architecture allows the model to have access to both past and future context, improving its ability to understand and predict complex temporal dependencies in the data[15].

Like self-attention mechanism in Transformers, the multi-head attention mechanism enhances the model's ability to focus on different feature subspaces. It helps the model identify and prioritize critical information when processing complex time series data.

(5) Weighted Mean Squared Error Loss Function

To place greater emphasis on the accuracy of predictions during critical dates such as holidays, this study customizes a Weighted Mean Squared Error (Weighted MSE) loss function defined as follows:

$$\text{Loss} = \frac{1}{N} \sum_{i=1}^N w_i (y_i - \hat{y}_i)^2 \quad (13)$$

Where w_i is the sample weight, y_i is the true value, and $\hat{y}_i$ is the predicted value. Sample weights are assigned based on whether the sample date is a holiday and its proximity to the holiday, aiming to improve the model's predictive performance on critical dates.

C. Data Partitioning

The dataset is divided into training and testing sets, with the training set covering the years 2012 to 2013 and the testing set comprising data from 2014.

D. Input Data

The model's inputs encompass a variety of features, detailed in Table II.

TABLE II THE FEATURES OF MODEL

Feature Category	Feature Name	Description
Weather Features	Max_Temp	Maximum temperature (°C)
	Min_Temp	Minimum temperature (°C)
	Avg_Temp	Average temperature (°C)
	Humidity	Relative humidity (%)
	Rainfall	Rainfall (mm)
Temporal Features	Year	Year
	Month	Month
	Day	Day
	DayOfWeek	Day of the week (0-6, 0 represents Monday)
	Season	Season code (0: Spring, 1: Summer, 2: Autumn, 3: Winter)
	IsWeekend	Whether it is a weekend (1: Yes, 0: No)
	DayOfMonth	Day of the month
	WeekOfYear	Week number
Holiday Features	IsHoliday	Whether it is a holiday (1: Yes, 0: No)
	IsWorkWeekend	Whether it is a compensatory working day (1: Yes, 0: No)
	IsSummerVacation	Whether it is summer vacation (1: Yes, 0: No)

Feature Category	Feature Name	Description
	IsWinterVacation	Whether it is winter vacation (1: Yes, 0: No)
	HolidayDuration	Duration of the holiday (days)
	DaysToSpringFestival	Days to the Spring Festival
	HolidayPeriod	Holiday period (0: Normal day, 1-5: Different stages)
Interaction Features	Temp_Holiday_Interaction	Interaction between average temperature and holiday
	Temp_SpringFestival_Interaction	Interaction between average temperature and days to Spring Festival
Lag and Rolling Statistical Features	Load_lag_1 to Load_lag_21	Load lags from 1 to 21 days
	Load_holiday_lag_1 to Load_holiday_lag_21	Load lags during holiday cycles
	Load_rolling_mean_7, 14, 30	Rolling mean of load for 7, 14, and 30 days
	Load_holiday_rolling_mean_7, 14, 30	Rolling mean of load during holiday cycles
Temperature and Load Variation Features	Temp_Change_Rate	Temperature change rate
	Temp_Rolling_Std	Rolling standard deviation of temperature
	Load_Change_Rate	Load change rate
	Load_Rolling_Std	Rolling standard deviation of load
	Load_Rolling_Max	Rolling maximum load
	Load_Rolling_Min	Rolling minimum load

IV. LOAD FORECASTING

Based on the aforementioned features, the deep learning model developed in this study was trained multiple epochs and achieved excellent forecasting performance in the test set. The specific training process and model performance are detailed below:

A. Model Training

1) Optimizer: Utilized the AdamW optimizer [16], defined by the following equation:

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{v_t} + \epsilon} (m_t + \lambda \theta_t) \quad (14)$$

2) Where η is the learning rate, m_t and v_t are the first and second moment estimates, respectively, and λ is the weight decay parameter.

3) Learning Rate Scheduler: Employed a Cosine Annealing learning rate scheduler [17], defined as:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\frac{T_{\text{cur}}}{T_i} \pi\right) \right) \quad (15)$$

4) Where T_{cur} is the current epoch number, T_i is the period length, and $\eta_{\min}$ and $\eta_{\max}$ are the minimum and maximum learning rates, respectively.

5) Gradient Clipping: Applied gradient clipping to prevent gradient explosion and enhance training stability.

6) Early Stopping: Implemented an early stopping mechanism with a patience of 10 epochs, allowing training to halt if the validation loss showed no improvement over 10 consecutive epochs.

B. Model Evaluation

(1) Evaluation Metrics

- Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (16)$$

- Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (17)$$

- Mean Absolute Percentage Error (MAPE):

$$\text{MAPE} = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (18)$$

(2) Experimental Results

The experimental results for Region A are summarized in Table III. Figures 2, 3, and 4 illustrate the training and validation loss curves, the progression of MAE, RMSE, and MAPE over time, and the comparison between predicted and actual load, respectively. Note that the metrics shown in Figure 4 represent the model's performance on normalized data during training, while the final results in Table III are inverse-transformed to their original scale for practical interpretation.

TABLE III MODEL EVALUATION METRICS FOR REGION A

Metric Name	Value
Test MAE	160.20 MW
Test RMSE	221.75 MW
Test MAPE	2.10%

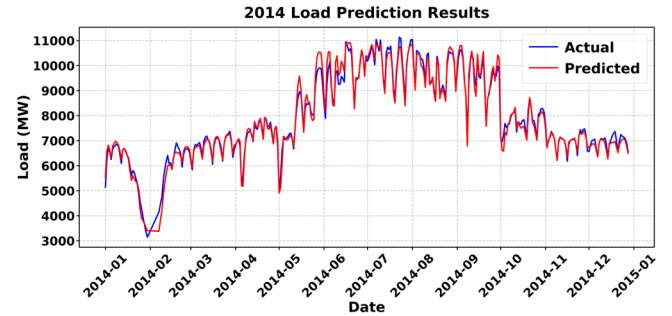


Figure 2: Predicted vs Actual Load for Region A

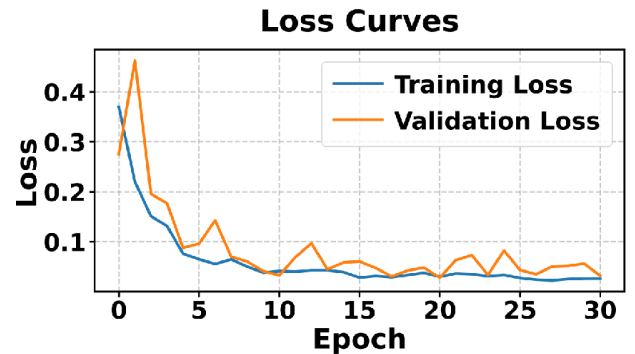


Figure 3: Training and Validation Loss Curve for Region A

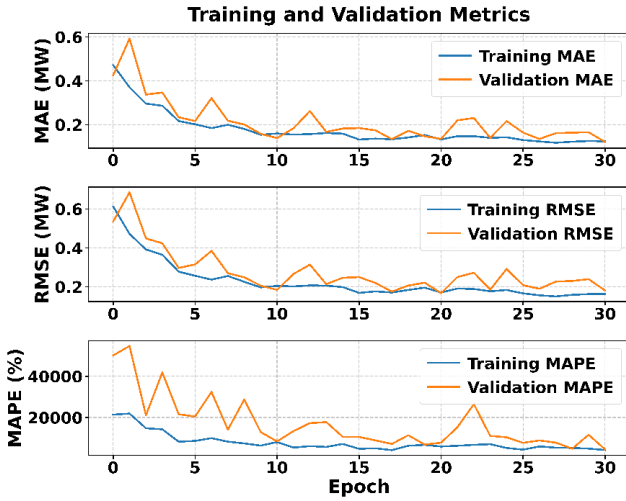


Figure 4: MAE, RMSE, and MAPE over Time for Region A

The training process demonstrated stable convergence, as evidenced by the gradual decrease in both training and validation losses, with the training loss reaching 0.0299 and validation loss reaching 0.0280.

(3) Forecast Assessment for an Additional Region

In order to further verify the generalization ability and applicability of the proposed model, this study applied the model to another Region B, power load data from 2012 to 2014 for prediction evaluation.

In the dataset of Region B, the same data preprocessing and feature engineering steps were used as in Region A. The model architecture and training parameters are kept consistent to ensure fairness and consistency of evaluation.

The experimental results for Region B are summarized in Table IV. Figures 5, 6, and 7 illustrate the training and validation loss curves, the progression of MAE, RMSE, and MAPE over time, and the comparison between predicted and actual load, respectively.

TABLE IV MODEL EVALUATION METRICS FOR REGION B

Metric Name	Value
Test MAE	190.39 MW
Test RMSE	318.44 MW
Test MAPE	3.45 %

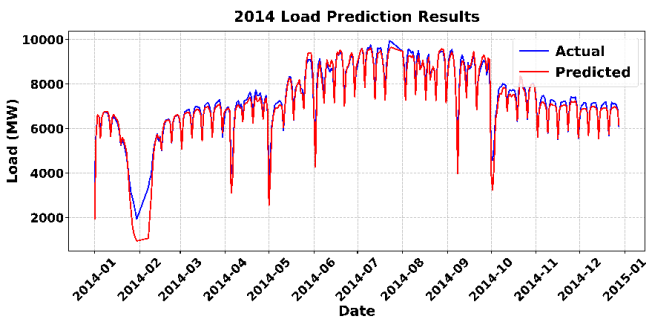


Figure 5: Predicted vs Actual Load for Region B

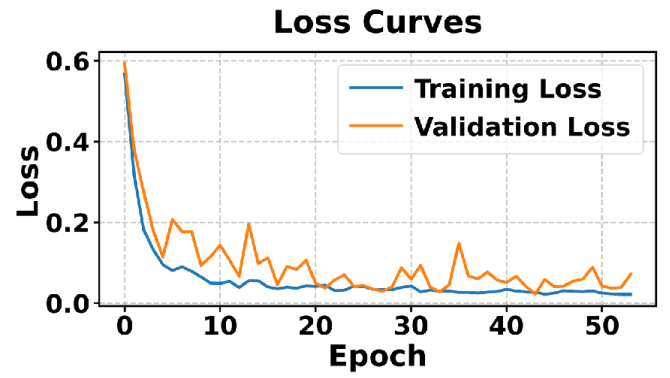


Figure 6: Training and Validation Loss Curve for Region B

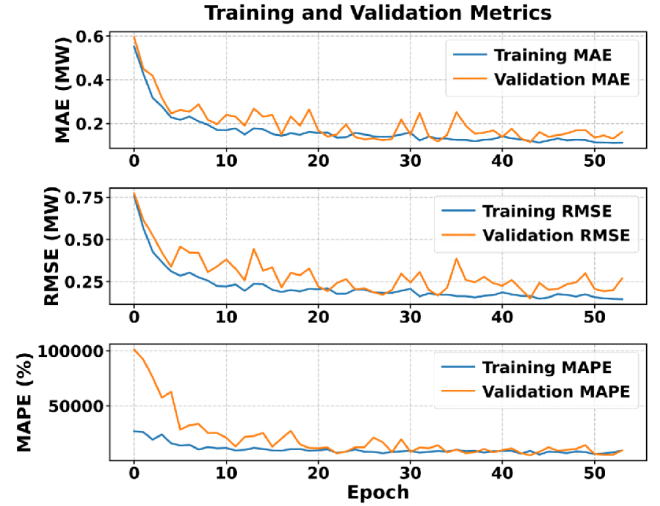


Figure 7: MAE, RMSE, and MAPE over Time for Region B

These results indicate that the model maintains robust forecasting performance in a different geographical area. Although the error metrics are slightly higher compared to those obtained in Region A, the model still demonstrates reliable predictive capability. The increased errors in Region B may be attributed to factors such as greater variability in power load patterns, differing climatic conditions, or unique regional demand characteristics. The model's consistent performance across both regions emphasizes its robust generalizability and suitability for a variety of power systems. The successful validation on a secondary region further confirms its reliability and adaptability, demonstrating its significant potential for practical applications in power load forecasting.

V. CONCLUSION

This study proposes a deep learning framework for power load forecasting that combines Transformer encoders, MambaBlock modules, and Bidirectional Long Short-Term Memory (Bi-LSTM) networks. By incorporating extensive feature engineering, particularly the meticulous consideration of holidays and compensatory workdays, the model demonstrates a strong ability to capture complex time series patterns and handle load variations during special dates. Experimental results show that the model achieves exceptional forecasting accuracy in Region A while maintaining robust performance in Region B. These findings underscore its adaptability and practicality for application across various regional power systems.

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